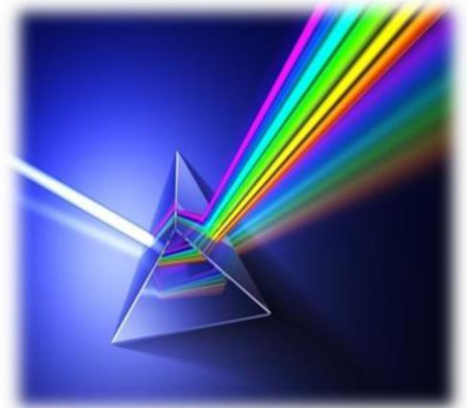


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II PUC PHYSICS

Model Paper - 01

2025 - 26

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Arise! Awake! and Stop not until The Goal is Reached

- Swami Vivekananda

II PUC – PHYSICS (33)

Time: 3 Hours 15 Min

Max. Marks: 70

No of questions: 45

General Instructions:

1. All parts A to D are compulsory.
2. Part – A questions have to be answered in the first two pages of the answer-booklet. For Part – A questions, first written-answer will be considered for awarding marks.
3. Answers without relevant diagram / figure / circuit wherever necessary will not carry any marks.
4. Direct answers to numerical problems without relevant formula and detailed solutions will not carry any marks.

PART – A

- I. Pick the correct option among the four given options for ALL of the following questions: 15 x 1 = 15

01. The electric flux over a closed surface enclosing a charged particle is

$-\frac{q}{6\epsilon_0}$. The charge of the particle is

(A) $+\frac{q}{6}$ (B) $+6q$ (C) $-\frac{q}{6}$ * (D) $-6q$

02. Two capacitors of capacitance C_1 and C_2 ($C_1 \neq C_2$) are connected in parallel to an external source of potential difference V . The equation which is not applicable for the given combination is (Symbols have their usual meanings)

(A) $Q = Q_1 + Q_2$ (B) $V = V_1 + V_2$ * (C) $C = C_1 + C_2$ (D) $V_1 = V_2$

03. In case of metals, with increase in temperature
- (A) the conductivity decreases*
 - (B) the average speed of conduction electrons decreases
 - (C) the resistivity decreases
 - (D) the number of conduction electrons increases
04. A galvanometer can be converted into a voltmeter by connecting:
- (A) a low resistance in series with it
 - (B) a high resistance in parallel with it
 - (C) a low resistance in parallel with it
 - (D) a high resistance in series with it*
05. In the following table, Column-I is the list of magnetic property and Column-II is the list of an item related to magnetic property. Identify the correct match.

Column – 1	Column – 2
(i) paramagnetism	(p) magnetic property common to all substances
(ii) diamagnetism	(q) domain formation
(iii) ferromagnetism	(r) sodium is an example for this magnetic property

(A) (i) – (q), (ii) – (r), (iii) – (p) (B) (i) – (r), (ii) – (q), (iii) – (p)

(C) (i) – (r), (ii) – (p), (iii) – (q)* (D) (i) – (p), (ii) – (q), (iii) – (r)

06. *Magnetic flux is*

(A) a scalar and its SI unit is Wb m^{-2}

(B) a scalar and its SI unit is Wb^*

(C) a vector and its SI unit is Wb m^{-2}

(D) a vector and its SI unit is Wb

07. *The resonance frequency of a series LCR circuit remains constant even if there is any variation in the value/values of*

(A) L (B) C (C) both L and R (D) R^*

08. *Electromagnetic waves are produced by:*

(A) charges in uniform motion (B) stationary charges

(C) accelerated charges*

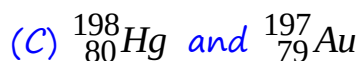
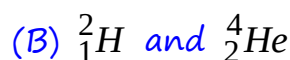
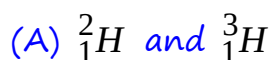
(D) steady current flowing through a conductor

09. *A real object is placed between the pole and principal focus of a concave mirror. The image formed is*

(A) virtual and magnified* (B) real and magnified

(C) virtual and diminished (D) real and diminished

10. The waves coming from two coherent sources overlap at a point to give maximum intensity. The phase difference between the superposing waves may be:
- (A) 3π (B) $6\pi^*$ (C) 5π (D) 7π
11. Identify the correct statement
- (A) Light waves exhibit diffraction but not sound waves
- (B) Sound waves exhibit diffraction but not light waves
- (C) Light waves exhibit diffraction but not polarization
- (D) Light waves exhibit both diffraction and polarization*
12. In a photoelectric experiment, the graph related to two quantities is a straight-line with slope equal to (h/e) . Then it is a graph of:
- (A) kinetic energy of photoelectrons versus intensity of incident light
- (B) photocurrent versus intensity of incident light
- (C) stopping potential versus frequency of incident light*
- (D) photocurrent versus collector plate potential
13. The total energy of an electron revolving in n^{th} orbit of hydrogen atom is the least for:
- (A) $n=3$ (B) $n=1^*$ (C) $n=2$ (D) $n=\infty$
14. An example for isobars is



15. When a pn -junction diode is reverse biased

(A) both the barrier height and width of depletion region increase*

(B) the barrier height increases but the width of depletion region decreases

(C) the barrier height decreases but the width of depletion region increases

(D) both the barrier height and the width of depletion region decrease

II. Fill in the blanks by choosing appropriate answer given in the bracket for ALL the following questions: 5 × 1 = 05

(90°, magnitude, magnetization, work function, 0°, polarity)

16. The net magnetic moment acquired per unit volume of a sample is called magnetization.

17. Lenz's law gives the polarity of the induced emf.

18. The phase difference between voltage and current in a pure resistive ac circuit is 0°.

19. The minimum energy required by an electron to escape from the metal surface is called work function.

20. If the angle of incidence for a ray of light in denser medium is equal to critical angle, then the angle of refraction in rarer medium is equal to 90°.

PART – B

III. Answer any FIVE of the following questions:

5 x 2 = 10

21. Define electric field at a point. What is meant by 'source charge'?

- The electric field at a point is defined as the force experienced by a unit test charge placed at that point.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

$\vec{E} \rightarrow$ electric field

$\vec{F} \rightarrow$ force experienced by the test charge q_0

- A source charge is an electric charge that produces electric field.

22. The electrostatic potential energy of a system of two like charges is positive. Explain.

- In order to form a system of two like charges, external work has to be done against the electrostatic repulsion between them. This is stored as the potential energy of the system.
- Electrostatic potential energy of two charge system is

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

As r and $q_1 q_2$ is positive for like charges, U is always positive for like charges.

23. Name the mobile charge carriers in metals. Write the expression for mobility in terms of average collision time (relaxation time) for charge carriers.

- The mobile charge carriers in metals are free electrons.
- Mobility of free electrons is

$$\mu = -\frac{e\tau}{m}$$

$\mu \rightarrow$ mobility of free electrons $e \rightarrow$ charge of the electron

$\tau \rightarrow$ relaxation time $m \rightarrow$ mass of the electron

24. What is the nature of the force between two parallel conductors carrying currents in the (a) same direction and (b) opposite direction?

- The force between the conductors is
 - attractive when the currents are in the same direction
 - repulsive when the currents are in the opposite direction

25. Calculate the magnetic energy stored in a coil of self-inductance 0.05 H carrying a current of 0.2 A .

- Energy stored in a coil,

$$U = \frac{1}{2} L I^2$$

$$U = \frac{1}{2} \times 0.05 \times 0.2^2$$

$$U = 1 \times 10^{-3} \text{ J}$$

26. Name the electromagnetic waves used in: (a) the radar systems of aircraft and (b) the laser assisted eye surgery (LASIK).

- The electromagnetic waves used in the
 - radar systems of aircraft are microwaves
 - laser assisted eye surgery (LASIK) are ultraviolet light

27. List out two differences between p-type and n-type semiconductors.

p-type semiconductor	n-type semiconductor
They are obtained by doping with trivalent impurities	They are obtained by doping with pentavalent impurities
Majority carriers are holes	Majority carriers are electrons
Minority carriers are electrons	Minority carriers are holes
Majority carriers move in valence band	Majority carriers move in conduction band
Minority carriers move in conduction band	Minority carriers move in valence band

28. Give any two limitations of Bohr's atomic model.

- The Bohr's atom model
 - is applicable only for hydrogen and hydrogen like atoms
 - could not explain the relativistic variation of mass of electron
 - could not explain the fine structure of spectral lines
 - could not explain the relative intensities of spectral lines
 - could not explain the splitting of spectral lines in a magnetic field (Zeeman effect) or in an electric field (Stark effect)

PART – C

IV. Answer any FIVE of the following questions:

5 x 3 = 15

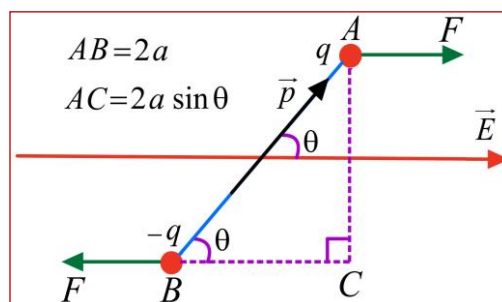
29. Derive an expression for torque acting on an electric dipole placed in a uniform electric field.

- Consider an electric dipole of dipole moment $p = q \times 2a$ placed in a uniform electric field $\vec{E}$.

At any instant, let θ be the angle between $\vec{p}$ and $\vec{E}$.

- Force on the charge $+q$ is

$$\vec{F} = q\vec{E} \quad \text{along } \vec{E}$$



Force on the charge $-q$ is

$$\vec{F} = q\vec{E} \quad \text{opposite to } \vec{E}$$

These two equal and unlike parallel forces constitute a couple.

- The magnitude of the torque is given by

torque = force on either charge $\times$ arm of the couple

$$\tau = F \times AC \quad \left[\sin \theta = \frac{AC}{AB} = \frac{AC}{2a} \Rightarrow AC = 2a \sin \theta \right]$$

$$\tau = qE \times 2a \sin \theta$$

$$\tau = p E \sin \theta$$

- In the vector form, $\vec{\tau} = \vec{p} \times \vec{E}$

$\vec{\tau}$ is perpendicular to the plane containing both $\vec{p}$ and $\vec{E}$.

30. What is a capacitor? Mention any two factors on which capacitance of a parallel plate capacitor depend.

- A capacitor is a device used for storing electric charges and electrical energy.
- Capacitance of parallel plate capacitor is given by

$$C = \frac{K \epsilon_0 A}{d}$$

- Capacity of a parallel plate capacitor depends on
 - area of the plates (A)

- separation between the plates (d)
- dielectric constant of the medium in between the plates (K)

31. In a chamber, a uniform magnetic field of $6.5 \times 10^{-4} \text{ T}$ is maintained. An electron is shot into the field with a speed of $4.8 \times 10^6 \text{ m s}^{-1}$ normal to the field. Determine the radius of the circular orbit. Given: $e = 1.6 \times 10^{-19} \text{ C}$ and $m_e = 9.1 \times 10^{-31} \text{ kg}$.

- Radius of the circular orbit of electron,

$$r = \frac{m_e v}{q B}$$

$$r = \frac{9.1 \times 10^{-31} \times 4.8 \times 10^6}{1.6 \times 10^{-19} \times 6.5 \times 10^{-4}}$$

$$r = 4.2 \times 10^{-2} \text{ m}$$

32. Mention any three properties of magnetic field lines.

Properties of magnetic field lines

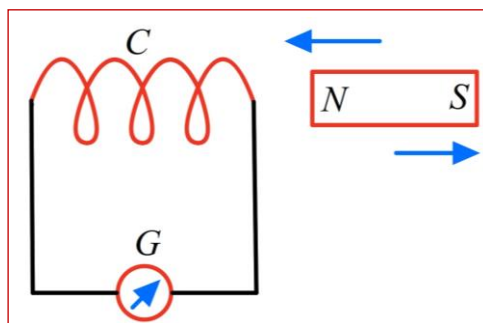
- Each magnetic field line forms a closed loop
- The tangent drawn to the magnetic field line at any point gives the direction of the magnetic field at that point
- Magnetic field lines never intersect each other
- In a uniform magnetic field, the magnetic field lines are straight and parallel

- They pass through the magnetic materials
- They contract longitudinally and widen laterally
- They are crowded where the magnetic field is strong and less crowded where the magnetic field is weak

33. Describe the coil and bar magnet experiment to demonstrate the phenomenon of electromagnetic induction.

- C is a coil connected to a sensitive galvanometer G.

NS is a bar magnet.



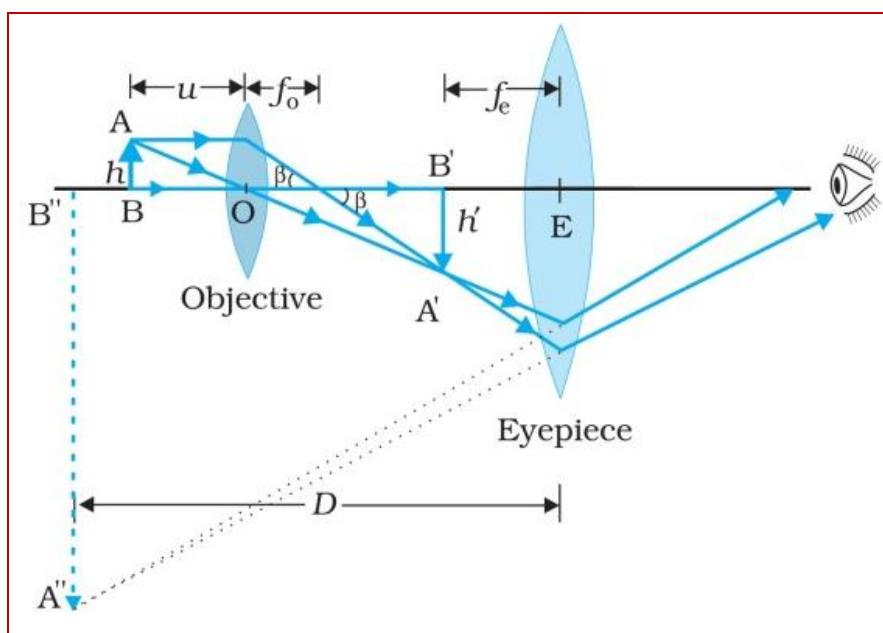
- When a bar magnet is moved towards the coil, galvanometer shows a momentary deflection and shows opposite deflection when the magnet is moved away from the coil.
- When there is no relative motion between the coil and the magnet, the galvanometer shows no deflection.

When the relative motion is brought about faster, the deflection is large.

- When there is a relative motion between the coil and the magnet, the magnetic flux linking with the coil changes and hence an emf is induced in the coil.

When the relative motion is brought about faster, the rate of change of magnetic flux with the coil increases. This increases the induced emf.

34. Draw the ray diagram for the formation of image by a compound microscope. Write the expression for linear magnification produced by the objective of compound microscope in terms of its tube length.



- When the image formed at D ,

$$M = \frac{L}{-f_o} \left(1 + \frac{D}{f_e} \right)$$

$L = (f_o + f_e) \rightarrow$ length of the microscope

$f_o \rightarrow$ focal length of the objective

$f_e \rightarrow$ focal length of the eyepiece

- When the image formed at infinity,

$$M = \frac{L}{-f_o} \times \frac{D}{f_e}$$

35. What are matter waves? Explain the de Broglie hypothesis of matter waves.

- The waves associated with material particles in motion are called de-Broglie waves or matter waves.
- According to de-Broglie, just as light exhibits both wave and particle properties (wave-particle duality), material particle also possesses dual nature under suitable conditions.
- A material particle of mass m moving with a velocity v associated with a wavelength λ . This is called de-Broglie wavelength.

$$\lambda = \frac{h}{m v} = \frac{h}{p}$$

$h \rightarrow$ Planck's constant

$p \rightarrow$ momentum of the particle

This hypothesis was later experimentally confirmed by the Davisson-Germer electron diffraction experiment.

36. Differentiate between nuclear fission and nuclear fusion.

Nuclear fission	Nuclear fusion
It is a process in which a heavy nucleus splits into two lighter nuclei	It is a process in which two lighter nuclei combine to form a single heavier nucleus

Projectile is necessary	Projectile is not necessary
Mass defect per reaction is about 0.2 u	Mass defect per reaction is about 0.02 u
Energy per reaction is about 200 MeV	Energy per reaction is about 20 MeV
Energy per unit mass is less	Energy per unit mass is more
It does not require high temperature	It requires high temperatures (about 10^8 K)
It can be controlled	It cannot be controlled
It produces radioactive waste	It does not produce radioactive waste

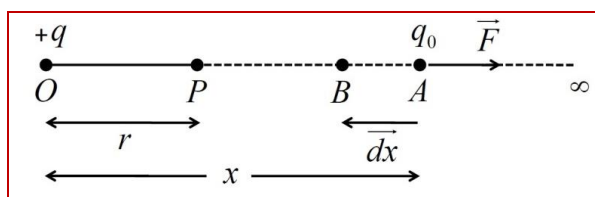
PART – D

V. Answer any THREE of the following questions:

3 x 5 =15

37. Derive an expression for electric potential at a point due to a point charge.

- Consider a point charge q placed at O in free space. Let P be a point at a distance r from the charge q .



- From Coulomb's law, the magnitude of the force on the test charge q_0 placed at A is

$$F = \frac{1}{4 \pi \epsilon_0} \frac{q q_0}{x^2} \quad \vec{F} \text{ acts away from the charge } q$$

- The work done in moving the test charge q_0 from A to B through a small displacement $d\vec{x}$ against the electrostatic force is

$$dW = F dx \cos 180^\circ = - F dx$$

- The total work done in moving the test charge q_0 from infinity to the point P is

$$W = \int dW = \int_{\infty}^r - F dx$$

$$W = \int_{\infty}^r - \frac{1}{4 \pi \epsilon_0} \frac{q q_0}{x^2} dx$$

$$W = \frac{1}{4 \pi \epsilon_0} q q_0 \int_{\infty}^r - \frac{1}{x^2} dx$$

$$W = \frac{1}{4 \pi \epsilon_0} q q_0 \left[\frac{1}{x} \right]_{\infty}^r$$

$$W = \frac{1}{4 \pi \epsilon_0} q q_0 \left[\frac{1}{r} - \frac{1}{\infty} \right]$$

$$W = \frac{1}{4 \pi \epsilon_0} \frac{q q_0}{r}$$

- Electric potential at the point P is

$$V = \frac{W}{q_0}$$

$$V = \frac{1}{4 \pi \epsilon_0} \frac{q}{r}$$

- The potential V is positive, if q is positive

The potential V is negative, if q is negative

38. Assuming the expression for drift velocity, derive the expression for conductivity of a material $\sigma = \frac{n e^2 \tau}{m}$, where symbols have their usual meaning.

- Consider a conductor of length l having uniform cross sectional area A .

Volume of the conductor $= Al$

- Total number of free electrons in the conductor $= n Al$

$n \rightarrow$ free electron density of the conductor

- Total free electron charge in the conductor $= q = e n Al$

$e \rightarrow$ charge of each free electron

- When the potential difference is applied across the conductor, an electric field is produced in the conductor. Due to this electric field, the electrons acquire a drift velocity v_d .

- The time t taken by this charge q to cross the conductor is

$$t = \frac{l}{v_d}$$

- Electric current I in the conductor is

$$I = \frac{q}{t} \quad \text{or} \quad I = \frac{e n A l}{\frac{l}{v_d}}$$

$$I = n e A v_d \quad \dots (1)$$

- But drift velocity of electrons is

$$v_d = \frac{e E \tau}{m} \quad \dots (2)$$

$\tau \rightarrow$ relaxation time $m \rightarrow$ mass of the electron

- Substituting (2) in (1),

$$I = n e A \frac{e E \tau}{m} \quad \text{or} \quad I = \frac{n e^2 \tau E A}{m}$$

$$\frac{I}{A} = \left(\frac{n e^2 \tau}{m} \right) E$$

$$j = \left(\frac{n e^2 \tau}{m} \right) E \quad \dots (3) \quad \frac{I}{A} = j \rightarrow \text{current density}$$

- From Ohm's law,

$$j = \sigma E \quad \dots (4)$$

$\sigma \rightarrow$ electrical conductivity

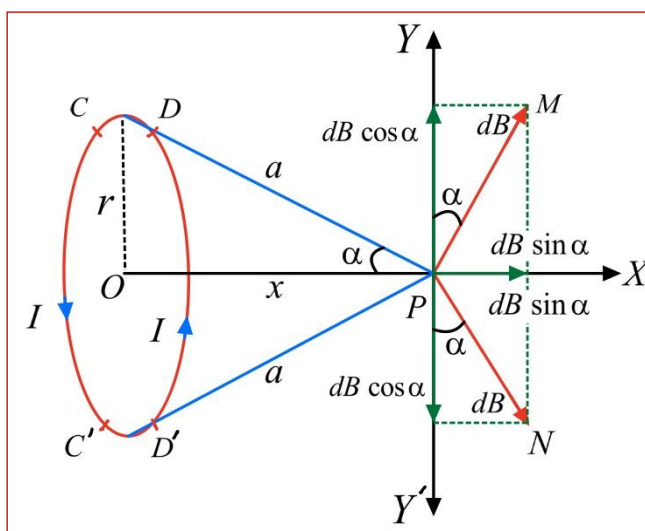
- On comparing (3) and (4),

$$\sigma = \frac{n e^2 \tau}{m}$$

39. Derive an expression for the magnetic field at a point on the axis of a circular current loop using Biot-Savart's law.

- Consider a current loop of radius r carrying current I in the direction as shown in the figure.
- Let the plane of the loop be perpendicular to the plane of the paper.

Let P be a point on its axis at a distance x from the centre O of the current loop.



The entire loop is assumed to be divided into a large number of small current elements each of length dl .

- Consider one such current element $CD = I \vec{dl}$ of the loop.

According to the Biot – Savart's law, the magnitude of magnetic field $\vec{dB}$ at the point P due this current element is

$$dB = \left(\frac{\mu_0}{4\pi} \right) \frac{I dl \sin 90^\circ}{a^2}$$

$$dB = \left(\frac{\mu_0}{4\pi} \right) \frac{I dl}{a^2} \quad \text{along PM}$$

- The magnetic field dB along PM can be resolved into its rectangular components
 - $dB \cos \alpha$ along PY (perpendicular to the axis of loop)
 - $dB \sin \alpha$ along PX (parallel to the axis of loop)
- For any two diametrically opposite current elements of the loop,
 - the components perpendicular to the axis of the loop will be equal and opposite and hence they will cancel out
 - the components parallel to the axis of the loop will be equal and along the same direction and hence they get added up
- The total magnetic field at the point P due to the loop is

$$B_A = \int dB \sin \alpha$$

But $dB = \left(\frac{\mu_0}{4\pi} \right) \frac{I dl}{a^2}$ and $\sin \alpha = \frac{r}{a}$

$$B_A = \int \left(\frac{\mu_0}{4\pi} \right) \frac{I dl}{a^2} \frac{r}{a}$$

$$B_A = \left(\frac{\mu_0}{4\pi} \right) \frac{I r}{a^3} \int dl$$

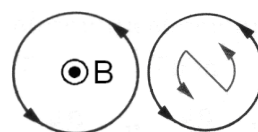
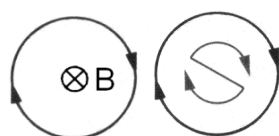
$$\int dl = 2\pi r \rightarrow \text{total length of all the current elements}$$

From the figure, $a = (r^2 + x^2)^{1/2}$

$$B_A = \left(\frac{\mu_0}{4\pi} \right) \frac{I r}{(r^2 + x^2)^{3/2}} \times 2\pi r$$

$$B_A = \left(\frac{\mu_0}{4\pi} \right) \frac{2\pi I r^2}{(r^2 + x^2)^{3/2}}$$

This field is along the axis of the loop and into the loop if the current is in the clockwise direction and away from the loop if the current is in the anticlockwise direction.

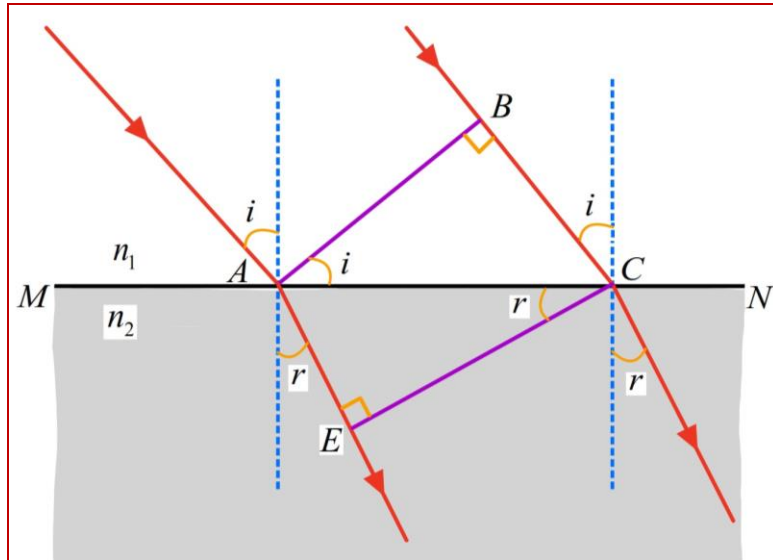


40. (a) What is a wavefront? (b) Derive Snell's law for a plane wave undergoing refraction using Huygens principle. (1+4)

- The locus of all points oscillating in the same phase is called a wavefront.
- Consider a plane surface MN separating a rarer medium of refractive index n_1 from a denser medium of refractive index n_2 .

Let v_1 and v_2 be the velocity of light in rarer and denser medium respectively.

Then $n_2 = \frac{v_1}{v_2} \dots (1)$



- Consider a plane wavefront AB incident on refracting surface at an angle i . Let the secondary wavelet from B strike the surface at C in time τ .

Then $BC = v_1 \tau$

- In the same time τ , the secondary wavelet from A will travel the distance $AE = v_2 \tau$ in the denser medium. The tangential surface CE represents the refracted wavefront. Let r be the angle of refraction.
- From the triangles BAC and ECA ,

$$\frac{\sin i}{\sin r} = \frac{\frac{BC}{AC}}{\frac{AE}{AC}} = \frac{BC}{AE} = \frac{v_1 \tau}{v_2 \tau}$$

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} \quad \dots (2)$$

- From (1) and (2),

$$\frac{\sin i}{\sin r} = {}_1n_2$$

Snell's law of refraction

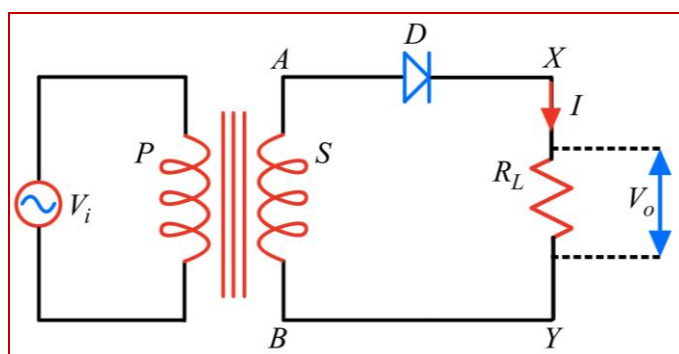
41. What is a half-wave rectifier? Draw a neat labelled circuit diagram and input-output waveforms for half-wave rectifier. Explain the working of half-wave rectifier.

Half-wave rectifier

- Half-wave rectifier is one in which there exists the rectified output for only one-half cycle of input AC.

Working of Half-wave rectifier

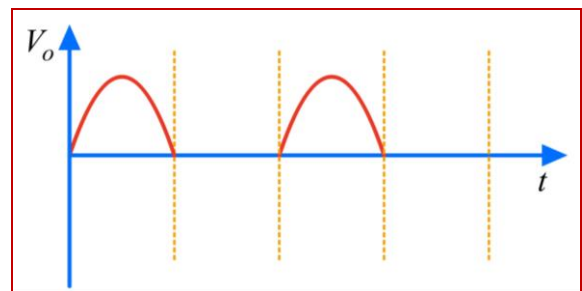
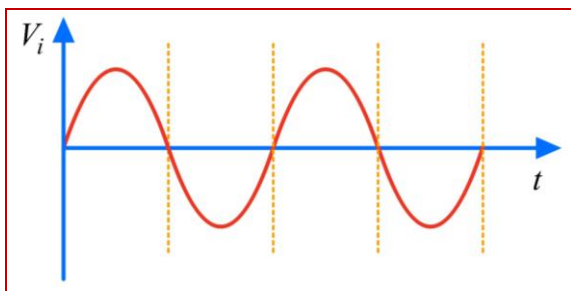
- The half wave rectifier circuit is as shown in the figure.



- AC voltage to be rectified is applied to the primary P of the transformer. It induces a voltage in the secondary S of the transformer.
- During the positive half cycle, A is positive. The diode D is forward biased and conducts. The current I flows through R_L from X to Y. Hence a voltage is developed across R_L .

During the negative half cycle, A is negative. The diode D is reverse biased and does not conduct. No current flows through R_L .

- This process occurs periodically. The output of the rectifier is not steady but pulsating DC.
- The input and the output voltage waveforms of a half wave rectifier are as shown in the figure.



VI. Answer any TWO of the following questions:

2 x 5 = 10

42. A positively charged spherical conductor of radius 0.1 m produces an electric field of $1.8 \times 10^3 \text{ N C}^{-1}$ at a point P distant 0.2 m from its centre. Calculate the magnitude of the charge present on the spherical conductor. Also find the magnitude of new charge produced on the same spherical conductor if 5×10^{10} extra electrons are removed from it.

- $R = 0.1 \text{ m}$, $E = 1.8 \times 10^3 \text{ N C}^{-1}$, $r = 0.2 \text{ m}$, $N = 5 \times 10^{10}$, $q_1 = ?$, $q_{\text{new}} = ?$
- Electric field at P,

$$E = \frac{1}{4 \pi \epsilon_0} \frac{q_1}{r^2}$$

$$1.8 \times 10^3 = \frac{9 \times 10^9 \times q_1}{(0.2)^2}$$

$$q_1 = \frac{1.8 \times 10^3 \times (0.2)^2}{9 \times 10^9}$$

$$q_1 = 8 \times 10^{-9} \text{ C}$$

- Additional charge on the sphere on the removal of electrons,

$$q_2 = N e$$

$$q_2 = 5 \times 10^{10} \times 1.6 \times 10^{-19}$$

$$q_2 = 8 \times 10^{-9} \text{ C}$$

- New charge on the sphere,

$$q_{\text{new}} = q_1 + q_2$$

$$q_{\text{new}} = (8 + 8) \times 10^{-9}$$

$$q_{\text{new}} = 16 \times 10^{-9} \text{ C}$$

43. The terminal potential difference of a cell connected to an external resistance of 4Ω is 2.4 V . When the same cell is connected to an external resistance of 9Ω , the new terminal potential difference becomes 2.7 V . Find the emf and internal resistance of the cell.

- $R_1 = 4 \Omega$, $V_1 = 2.4 \text{ V}$, $R_2 = 9 \Omega$, $V_2 = 2.7 \text{ V}$, $E = ?$, $r = ?$
- Terminal potential difference of the cell,

$$V_1 = \frac{E R_1}{R_1 + r}$$

$$2.4 = \frac{E \times 4}{4 + r} \quad \text{or} \quad E = \frac{2.4(4 + r)}{4}$$

$$E = 2.4 + 0.6 r \quad \dots (1)$$

$$V_2 = \frac{E R_2}{R_2 + r}$$

$$2.7 = \frac{E \times 9}{9 + r} \quad \text{or} \quad E = \frac{2.7(9 + r)}{9}$$

$$E = 2.7 + 0.3 r \quad \dots (2)$$

- From (1) and (2),

$$2.4 + 0.6 r = 2.7 + 0.3 r$$

$$0.3 r = 0.3$$

$$r = 1 \Omega$$

- From (1),

$$E = 2.4 + 0.6 r$$

$$E = 2.4 + 0.6 \times 1$$

$$E = 3 V$$

44. A resistor of 10Ω and an inductor of self-inductance $0.05 H$ are connected in series to an ac source of $200 V, 50 Hz$. Find the impedance of the circuit and current in the circuit.

- $R = 10 \, \Omega$, $L = 0.05 \, H$, $V_{rms} = 200 \, V$, $f = 50 \, Hz$, $Z = ?$, $I_{rms} = ?$

- Inductive reactance,

$$X_L = 2 \pi f L$$

$$X_L = 2 \times 3.14 \times 50 \times 0.05$$

$$X_L = 15.7 \, \Omega$$

- Impedance,

$$Z = \sqrt{R^2 + X_L^2} \quad [X_C = 0]$$

$$Z = \sqrt{10^2 + 15.7^2}$$

$$Z = 18.61 \, \Omega$$

- Current in the circuit,

$$I_{rms} = \frac{V_{rms}}{Z}$$

$$I_{rms} = \frac{200}{18.61}$$

$$I_{rms} = 10.75 \, A$$

45. Double convex lenses are to be manufactured from a glass of refractive index 1.55 with both faces of the same radius of curvature. Find the radius of curvature required if the focal length is to be 20 cm. Also find the new focal length of the lens if it is immersed in water of refractive index 1.33.

- $n_g = 1.55$, $R_1 = R$, $R_2 = -R$, $f = 20 \text{ cm}$, $n_w = 1.33$, $R = ?$, $f_w = ?$
- Focal length of the lens in air,

$$\frac{1}{f_a} = (n_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{20} = (1.55 - 1) \left(\frac{1}{R} + \frac{1}{R} \right)$$

$$\frac{1}{20} = \frac{0.55 \times 2}{R}$$

$$R = 0.55 \times 2 \times 20$$

$$R = 22 \text{ cm}$$

- When immersed in water,

$$\frac{f_w}{f_a} = \frac{(n_g - 1)}{\left(\frac{n_g}{n_w} - 1 \right)}$$

$$\frac{f_w}{20} = \frac{(1.55 - 1)}{\left(\frac{1.55}{1.33} - 1 \right)} = \frac{0.55}{0.1654}$$

$$f_w = \frac{0.55 \times 20}{0.1654}$$

$$f_w = 66.51 \text{ cm}$$



GOVERNMENT OF KARNATAKA
KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD

SUBJECT: PHYSICS (33)

BLUEPRINT FOR 2025-26 II PUC

UNIT	CH NO	CHAPTER	NO. OF TEACHING HOURS	Marks	PART-A	PART-B	PART-C	PART-D
					1 MARK QUESTIONS NO OF Qs 20	2 MARK QUESTIONS NO OF Qs 8	3 MARK QUESTIONS NO OF Qs 8	5 MARK QUESTIONS NO OF Qs 9
I	1	Electric charges & Fields	12	11	1	1	1	1
II	2	Electrostatic potential and Capacitance	12	11	1	1	1	1
III	3	Current electricity	15	13	1	1		2
IV	4	Moving Charges and Magnetism	13	11	1	1	1	1
V	5	Magnetism and Matter	6	5	2		1	
	6	Electromagnetic Induction	8	7	2	1	1	
VI	7	Alternating Current	8	7	2			1
	8	Electromagnetic Waves	3	3	1	1		
VII	9	Ray Optics and Optical Instruments	11	10	2		1	1
VIII	10	Wave Optics	8	7	2			1
IX	11	Dual nature of Radiation and Matter	6	5	2		1	
	12	Atoms	4	3	1	1		
X	13	Nuclei	5	4	1		1	
	14	Semiconductor Electronics	9	8	1	1		1
TOTAL	NO OF QUESTIONS		120	105	20/20	05/08	05/08	05/09
	MARKS				20/20	10/16	15/24	25/45

All the Very Best Stay Safe